



CADGCL: unsupervised retrieval of CAD models via boundary representations

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Abstract

With the widespread application of CAD technology in the industrial manufacturing sector, the efficient retrieval of target models has become a critical research topic. Despite the outstanding performance of traditional supervised retrieval methods, their reliance on large amounts of labeled data significantly limits practical applications. Data labeling is not only time-consuming and costly but also difficult to ensure accuracy and consistency. To tackle this issue, this paper introduces CADGCL, an unsupervised method for retrieving CAD models based on boundary representations. The proposed method transforms CAD models represented by boundary representations into B-rep attributed graphs that integrate geometric information and topological structures. Graph contrastive learning facilitates unsupervised CAD model retrieval. To overcome the limitations of traditional GCL methods in data augmentation and negative sampling, two novel strategies are introduced: an edge perturbation strategy based on Edge Betweenness Centrality and a negative sampling strategy based on the Beta Mixture Model. These strategies effectively improve the performance of contrastive learning. Experimental results show that the proposed method outperforms existing approaches in mAP and F1 scores under unsupervised scenarios, validating its potential for applications in industrial manufacturing.

Keywords CAD model retrieval · Unsupervised learning · Boundary representation · Contrastive graph learning

1 Introduction

CAD models are critical in industrial manufacturing and product design since they are the foundation for the entire production process. Studies show that approximately 75–80% of new part models reuse historical data through variant and adaptive design, while only about 20–25% of new parts require a complete redesign[8]. These data highlight the significant reuse value of CAD models in the design process, particularly in large-scale projects or complex product designs, where quickly retrieving suitable existing models is essential for improving design efficiency, reducing redundant work, and fostering innovation.

With the rapid development of deep learning, numerous methods have emerged that significantly improve the efficiency and accuracy of CAD model retrieval [8, 9, 12, 13, 20]. Most of these methods rely on supervised or semi-supervised frameworks, where labeled data guide the learning process. However, obtaining sufficient and reliable labels is both challenging and expensive. Data annotation is time-consuming and costly, and it is challenging to ensure accuracy and consistency in industrial design, where the diversity of CAD

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models and complexity of design requirements pose additional challenges.

In response to these challenges, there has been a growing focus on unsupervised learning methodologies for CAD model retrieval in recent years [14, 17, 30, 32]. However, existing unsupervised CAD model retrieval methods primarily focus on geometric representation formats such as point clouds and meshes, with relatively little emphasis on CAD model retrieval based on boundary representations (B-rep). In industrial design, B-rep is a widely used model representation method in CAD systems that can accurately describe complex geometric shapes and their topological characteristics[8]. B-rep includes geometric data and detailed topological descriptions of boundaries, faces, and edges, making it suitable for high-precision engineering design. However, unsupervised methods lack effective strategies for B-rep-based CAD model retrieval.

To address this gap, this paper proposes CADGCL, an unsupervised method for CAD model retrieval based on B-rep. In particular, we extract information from the STEP files of CAD models to construct B-rep attributed graphs and employ graph contrastive learning (GCL) to learn powerful embeddings without relying on extensive labeled data. GCL is particularly well-suited to this setting, as it naturally accommodates the relational structure and topology inherent in B-rep models, thereby enabling the capture of both geometry and topology and facilitating the learning of robust embeddings. Nonetheless, incorporating GCL into B-rep-based CAD retrieval introduces two key challenges. First, designing graph data augmentations that preserve essential semantic information: prior methods typically rely on random edge perturbations, which risk destroying the critical structure and thus diminishing semantic fidelity. Second, establishing unbiased negative sampling strategies: most existing approaches treat all remaining samples in a mini-batch as negative, which can inadvertently lead to false negatives (instances where negative samples share the same label as the positive), ultimately undermining the contrastive learning objective. To tackle these challenges, we introduce an edge perturbation strategy based on Edge Betweenness Centrality (EBC) and a negative sampling strategy based on the Beta Mixture Model (BMM), significantly enhancing the performance of contrastive learning. Experimental results demonstrate that the proposed unsupervised CAD model retrieval method based on B-rep achieves outstanding performance across multiple datasets. Under unsupervised scenarios, the method achieves superior mean Average Precision (mAP) and F1 scores compared to existing supervised and unsupervised methods, validating its practicality and effectiveness in industrial manufacturing. The primary contributions of this paper can be summarized as follows:

- This paper proposes an innovative unsupervised CAD model retrieval method, which converts CAD models into B-rep attributed graph representations and leverages the GCL framework to achieve efficient CAD model retrieval.
- To overcome the limitations of traditional GCL methods, this study presents an edge perturbation strategy based on the EBC and an unbiased negative sampling strategy based on the BMM, thus enhancing the performance of contrastive learning.
- The experimental results of two datasets show that the proposed method achieves state-of-the-art results and significantly exceeds existing unsupervised methods in terms of mAP and F1 scores.

2 Related work

2.1 Supervised CAD model classification and retrieval

As deep learning advances rapidly, many methods have been developed that greatly enhance the efficiency and accuracy of CAD model retrieval. Wu et al. [23] introduced 3D ShapeNets, which convert 3D models into voxel grid representations and utilize 3D convolutional neural networks to learn feature representations of 3D shapes, enabling effective 3D model retrieval. Maturana et al. [15] developed VoxNet, a real-time 3D object recognition method that employs sparse convolution to skip zero values in voxel grids, accelerating computation and achieving efficient model classification.

Additionally, methods for representing 3D geometric shapes using mesh structures have been widely researched. Mesh discretizes and approximates 3D models by connecting vertices, edges, and triangular faces, enabling efficient representation of objects' geometric shapes and topological structures. Hanocka et al.[7] proposed MeshCNN, which performs convolution operations on mesh edges and their neighboring edges. Dai and He et al.[2] introduced MEAN, an attention-based 3D mesh shape classification method that incorporates edge attention modules and combines global feature learning with local feature aggregation, significantly improving classification performance.

Point cloud representations can be directly obtained from 3D scanning data, avoiding information loss during voxelization and meshing processes while significantly reducing computational and storage costs. Qi et al.[18] proposed PointNet, which directly applies deep learning to point cloud data to achieve 3D classification and segmentation. Subsequently, Qi et al. [19] introduced PointNet++, which incorporates hierarchical feature learning into PointNet, enabling the capture of local structural features in point cloud data. Zhang and He et al.[31] proposed PointwiseNet, which achieves

efficient 3D point cloud classification and segmentation by learning low-level geometric and high-level semantic information point by point. Wang et al.[22] developed DGCNN, which dynamically constructs graph structures to learn features from point cloud data, allowing the network to adaptively capture local geometric information and thereby enhance point cloud classification and segmentation performance. Kim et al.[12] proposed a method for generating depth images to facilitate view-based shape retrieval of 3D CAD models from partial point clouds, effectively bridging the gap between incomplete input data and accurate retrieval results through robust representation learning.

In addition to point cloud representation, another standard 3D geometric representation method is boundary representation. B-rep is a widely used model representation method in CAD systems, describing the boundaries of 3D objects using geometric elements such as vertices, edges, and faces. Jayaraman et al.[9] proposed UV-net, which converts B-rep models into face adjacency graphs to learn geometric and topological information, thereby enabling efficient shape matching and feature learning. Hou et al.[8] proposed FuS-GCN, which leverages graph neural networks to extract geometric and topological features of CAD models. Integrating a self-attention mechanism and feature fusion into the graph pooling algorithm achieves superior classification and retrieval performance. Qin et al.[20] proposed VGNet, a multimodal feature extraction and fusion network designed for 3D CAD model retrieval. By integrating visual and geometric features, VGNet employs a dual-stream architecture that processes image and graph representations of CAD models. Lee et al.[13] proposed BRepGAT, a graph neural network designed to segment machining feature faces in B-rep models with high precision and robustness.

2.2 Unsupervised CAD model classification and retrieval

Unsupervised learning methods are increasingly applied in CAD model retrieval, particularly in scenarios with scarce labeled data, showcasing their unique advantages. By automatically learning feature representations of CAD models, unsupervised methods not only circumvent the high costs of manual labeling but also better adapt to large-scale unlabeled datasets. Pang et al.[17] proposed Point-MAE, an autoencoder-based framework that trains the model by masking parts of the point cloud data to automatically learn point cloud features. Zha et al.[30] introduced Point-FEMAE, which combines global and local branches to capture latent semantic features, enabling the learning of compact 3D point cloud representations. Liang et al.[14] proposed MeshCL, a contrastive learning method for 3D meshes that enhances geometric diversity, achieving robust feature representations

for geometric deformations and state-of-the-art performance in mesh classification and segmentation tasks.

Despite the significant progress achieved by the aforementioned methods in various CAD model retrieval tasks, they often rely heavily on large amounts of manually labeled data and face limitations in generalization ability and data diversity. In particular, supervised learning methods may encounter challenges such as training difficulties and insufficient feature extraction when dealing with large-scale unlabeled datasets. While autoencoder-based unsupervised learning methods lessen the reliance on labeled data, they frequently face challenges such as inadequate feature learning, high computational complexity, and unstable training. These issues have hindered the application of existing methods in large-scale CAD model repositories. This paper proposes an unsupervised CAD model retrieval method based on B-rep to address these shortcomings. The proposed method enables unsupervised CAD model retrieval by transforming B-rep models into graph representations and leveraging graph contrastive learning.

2.3 Graph contrastive learning

GCL is an unsupervised learning method based on graph data, which achieves self-supervised feature learning by designing positive and negative sample pairs, thereby efficiently capturing graph structural information. GraphCL[29] first proposed four graph data augmentation strategies, such as node dropping and edge perturbation, and utilized these strategies to generate positive and negative sample pairs. Building on GraphCL, JOAO[28] introduced adaptive and dynamic mechanisms to select graph data augmentation strategies, eliminating the cumbersome process of manual augmentation selection. AutoGCL[27] automatically generates multiple graph views, further reducing reliance on manual augmentation strategies while improving the adaptability of graph contrastive learning. SimGRACE[24] enhances data by introducing Gaussian noise perturbations to the model, simplifying traditional augmentation strategies in graph contrastive learning. DRGCL[10] optimizes the dimensionality selection process in graph contrastive learning by introducing learnable dimensionality principles and redundancy reduction constraints. GCLMI[26] maximizes the mutual information between each augmented view and the original graph while minimizing the mutual information between different augmented views, resulting in feature representations that are both informative and diverse. Tan et al.[21] proposed CIGCL, which preserves community structures during graph data augmentation. This method unifies topology and feature constraints by maximizing spectral changes, thereby enhancing the model's robustness against noise.

Although existing GCL methods have improved the feature learning capabilities of graph data to some extent, they

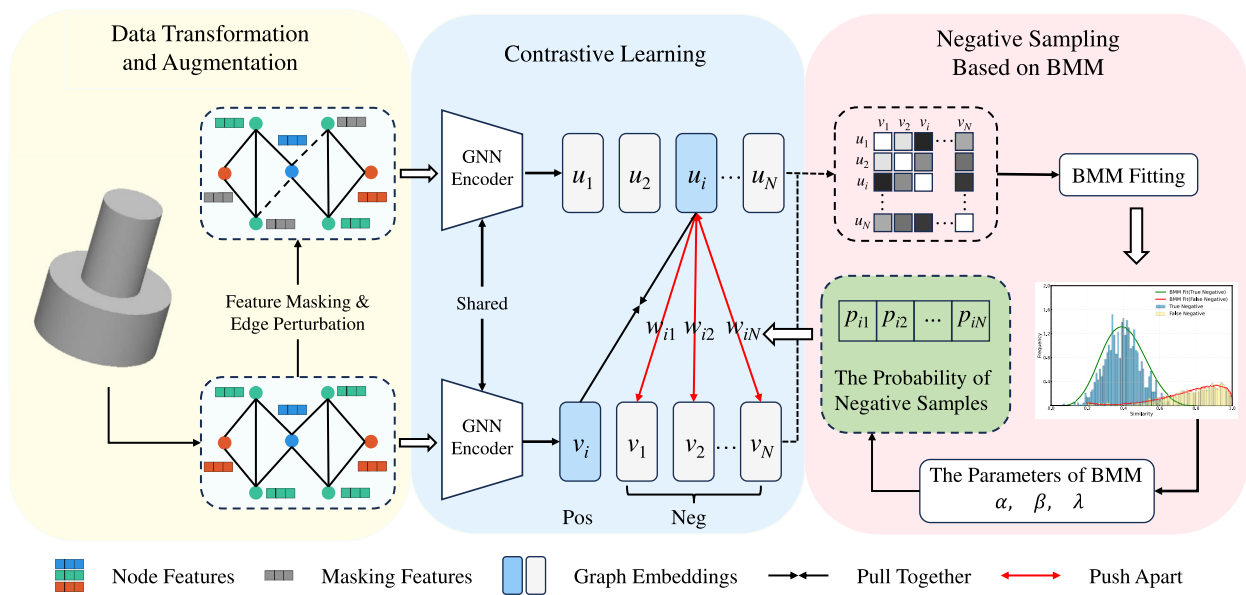


Fig. 1 Overview of our approach CADGCL. Our method consists of three components. Data Transformation and Augmentation: converting B-rep models from CAD models to B-rep attribute graphs and obtaining corresponding enhanced views through data augmentation. Contrastive Learning: using a GNN encoder to obtain graph embeddings of B-

rep attribute graphs and corresponding enhancement graphs in a batch, where positive pairs are pulled together and negative pairs are pushed apart. Negative Sampling Based on BMM: BMM estimates the probability of true and false negatives, and the estimated probabilities are used to improve the contrastive loss function

still face limitations in data augmentation and negative sampling. Traditional GCL methods generate augmented views through random edge perturbations, which inevitably disrupt the semantic information of the graphs, leading to suboptimal results. Furthermore, existing negative sampling methods often rely on random selection, which is prone to sampling bias and degrades model performance. This paper proposes an edge perturbation strategy based on EBC and an unbiased negative sampling method based on BMM to address these issues. These strategies aim to overcome the limitations of existing GCL methods and enhance the performance of the model.

3 Method

This section will provide a detailed introduction to our proposed unsupervised CAD model retrieval method CADGCL based on B-rep. Figure 1 shows the overall architecture of the model. Our proposed method consists of three main parts: *Data Transformation and Augmentation*, *Contrastive Learning*, and *Negative Sampling Based on BMM*. In the first stage, the input CAD model is converted into a B-rep attribute graph, and then the corresponding enhanced view is obtained through feature masking and edge perturbation based on EBC. In the second stage, a GNN encoder is used to obtain graph embeddings of B-rep attribute graphs and corresponding enhancement graphs in a batch, where positive

pairs are pulled together, and negative pairs are pushed apart. In the third stage, the parameters of the probability density function of BMM are fitted by the similarity between sample pairs. Then, the learned parameters are used to estimate the probabilities of true and false negatives, which are used as weights to improve the contrastive loss function.

3.1 Data transformation and augmentation

Traditional 3D model retrieval methods rely on manually designed geometric features, like bounding boxes, surfaces, and topological structures, which are limited in accuracy for complex models. The Standard for the Exchange of Product Model Data (STEP) [4] file serves as an intermediate exchange format for 3D CAD models, playing a crucial role in the geometric data exchange. In this paper, the B-rep attributed graphs are obtained by extracting information from the STEP files of 3D CAD models, as shown in Fig. 2.

In the B-rep attributed graph, nodes represent CAD model faces, and edges represent the line or curve segments between them. The B-rep attributed graph is structurally associated with the 3D CAD model, where greater distances between nodes indicate weaker structural dependencies in the CAD model. After obtaining the B-rep attributed graph, we represent the graph as $G = (V, E)$, where V denotes the set of N nodes and E represents the set of edges. The adjacency matrix of the graph is $A \in \{0, 1\}^{N \times N}$, and the node feature matrix is $x \in \mathbb{R}^{N \times d}$, where d is the feature vector of node v_i

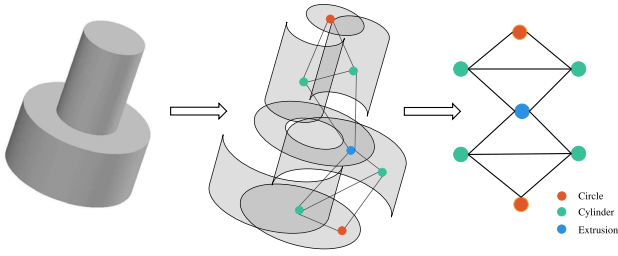


Fig. 2 CAD model and corresponding B-rep attribute graph

with a dimension of d . If there exists an edge E_{ij} between nodes v_i and v_j , then $A_{ij} = 1$.

Feature augmentation brings significant advantages to GNNs, especially in the context of combined topological and feature-level augmentation, which has been shown to greatly enhance GCL performance[33]. Inspired by this, we adopt a combined topological and feature augmentation approach in our graph data augmentation scheme. Specifically, unlike existing methods that achieve topological augmentation through simple random perturbations, we propose a topological augmentation strategy based on EBC. This approach preserves the critical structures of the original graph while uncovering and retaining hidden community structure information, providing rich information for learning more discriminative representations. On this basis, we further apply feature masking to enhance the learning of key information, significantly improving the precision of feature extraction.

Feature Masking: A mask vector $m \in \{0, 1\}^N$ is to randomly sampled to mask node features, setting the feature value of particular nodes to 0. Each element in the mask m is sampled from a Bernoulli distribution $\text{Ber}(1 - p_{\text{mask}})$, where the hyperparameter p_{mask} represents the feature drop rate. Therefore, the enhanced node feature matrix of the graph is $X' \in \mathbb{R}^{N \times d}$:

$$X' = [x_1 \circ m, x_2 \circ m, \dots, x_N \circ m], \quad (1)$$

Edge Dropping Based on EBC: Edge Betweenness Centrality[5] refers to the number of shortest paths between pairs of nodes in a graph that passes through a specific edge, where the shortest path is defined as the path with the minimum distance between any two nodes. EBC can be formalized as follows:

$$EBC(e) = \sum_{s, t \in V} \frac{\sigma(s, t | e)}{\sigma(s, t)}, \quad (2)$$

where $\sigma(s, t)$ represents the shortest paths between nodes s and t , and $\sigma(s, t | e)$ denotes the number of those paths that pass through edge e . Each edge is associated with its corresponding EBC value in the B-rep attributed graph. Intu-

itively, edges with high EBC often act as bridges between distinct communities, playing a pivotal role in maintaining global connectivity. Removing such edges suppresses potentially noisy inter-community links while reinforcing local structural consistency, thereby better reflecting the inherent modularity of CAD models. The resulting perturbed graph is denoted as $G' = (V, E')$, where $E' \in E$ represents the updated edge set after perturbation. In G' , intra-community connections become relatively denser, and inter-community interference is reduced. This topological augmentation preserves semantic coherence within local neighborhoods and facilitates the learning of more robust and discriminative representations in the contrastive learning framework.

3.2 Contrastive learning

After obtaining the b-rep graphs and corresponding enhanced views, we use a GNN encoder with shared parameters to obtain the corresponding graphs embedding. The node features and the topological relationships of edges from the B-rep attribute graph and its corresponding augmented view are input into a GNN encoder to obtain the embedding representation for each node. Subsequently, a readout function aggregates all node embeddings to generate the corresponding graph embeddings for the B-rep attribute graph and its augmented view. Typically, the readout function is implemented using average pooling:

$$\begin{aligned} H &= \text{GNN}(X, E), & Z &= \text{Readout}(H), \\ H' &= \text{GNN}(X', E'), & Z' &= \text{Readout}(H'), \end{aligned} \quad (3)$$

where H and H' denote the node embeddings, while Z and Z' represent the corresponding graph embeddings. Subsequently, two projection layers are used to map Z and Z' into distinct vector spaces:

$$U = \sigma(g(Z, \psi)) \quad V = \sigma(g(Z', \psi)), \quad (4)$$

where $\sigma(\cdot)$ denotes a nonlinear activation function. The projection head is a two-layer perceptron with a bias term ψ . The GNN model is trained using a contrastive learning approach with batch training, where embeddings from the same view (the B-rep attribute graph and its corresponding augmented view) are encouraged to be as similar as possible, while embeddings from different views are kept distant. In each batch, u_i and v_i represent the i th graph within the batch. The InfoNCE[16] loss function for a positive sample pair (u_i, v_i) is:

$$\begin{aligned} l(u_i, v_i) &= -\log \left(\frac{\phi(u_i, v_i)}{\sum \phi(u_i, v_k) + \sum_{k \neq i} \phi(u_i, u_k)} \right), \\ \phi(u, v) &= \exp \left(\frac{\text{sim}(u, v)}{\tau} \right), \end{aligned} \quad (5)$$

where τ is the temperature parameter, and $\text{sim}(\cdot)$ is the cosine similarity between the two views. The overall objective of the contrastive loss is to maximize the average mutual information of all positive sample pairs:

$$\mathcal{L}_C = \frac{1}{2N} \sum_{i=1}^N (l(u_i, v_i) + l(v_i, u_i)) \quad (6)$$

3.3 Negative sampling based on BMM

In GCL, the selection of negative samples is critical to the quality of feature learning. Traditional negative sampling methods often rely on random selection, which may introduce bias in negative sampling and lead to false negatives. False negatives refer to positive samples that are incorrectly labeled as negative samples. Such samples interfere with the model's learning process during training and adversely affect retrieval accuracy. Therefore, accurately distinguishing true negatives from false negatives is essential. Inspired by ProGCL[25], we propose a negative sampling strategy based on the BMM. In graph contrastive learning, these false negatives typically exhibit higher similarity scores with the anchor than true negatives, resulting in a bimodal distribution of similarity values across negative samples. To model this phenomenon, we adopt a BMM, as the Beta distribution is well-suited for capturing continuous similarity values within the bounded interval[0, 1]. By fitting a two-component BMM, we estimate the posterior probability that each negative sample belongs to the true negative component. These probabilities are subsequently used to reweight the contribution of each negative sample in the contrastive loss, effectively mitigating the impact of false negatives and enhancing the robustness of the learned representations. The probability density function (PDF) of the Beta distribution [6, 11] is:

$$p(s|\alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} s^{\alpha-1} (1-s)^{\beta-1}, \quad (7)$$

where α and β are the parameters of the Beta distribution, and $\Gamma(\cdot)$ is the Gamma function. We model the distributions of true and false negatives by fitting a two-component Beta Mixture Model. The probability density function of the Beta Mixture Model on the similarity matrix s is:

$$p(s) = \sum_{c=1}^2 \lambda_c p(s|\alpha_c, \beta_c), \quad (8)$$

where λ_c are the mixture coefficients, The subscript c indicates which Beta Mixture Model's parameters the sample is used to fit. In practice, we iteratively update the parameters of the Beta Mixture Model using the Expectation-Maximization

(EM) algorithm[3]. In the E-step, we fix the parameters of the Beta Mixture Model and then use the Bayes rule to compute the responsibility of each sample for each distribution, which is the probability that a sample belongs to either a true negative or a false negative:

$$p(c|s) = \frac{\lambda_c p(s|\alpha_c, \beta_c)}{\sum_{j=1}^2 \lambda_j p(s|\alpha_j, \beta_j)}, \quad (9)$$

then, we calculate the weighted average and variance:

$$\begin{aligned} \mu_c &= \sum_{i=1}^B \eta_i s_i, \quad \sigma_c^2 = \sum_{i=1}^B \eta_i (s_i - \mu_c)^2, \\ \eta_i &= \frac{p(c | s_i)}{\sum_{i=1}^B p(c | s_i)} \end{aligned} \quad (10)$$

where B represents the total number of samples in the batch. In the M-step, we use the method of moments to estimate the parameters of the Beta Mixture Model:

$$\begin{aligned} \alpha_c &= \mu_c \left(\frac{\mu_c(1 - \mu_c)}{\sigma_c^2} - 1 \right), \quad \lambda_c = \frac{1}{B} \sum_{i=1}^B p(c | s_i), \\ \beta_c &= (1 - \mu_c) \left(\frac{\mu_c(1 - \mu_c)}{\sigma_c^2} - 1 \right), \end{aligned} \quad (11)$$

once the model is fitted, we can calculate the probability of each negative sample being a true negative using the posterior probability:

$$p(c|s) = \frac{\lambda_c p(s|\alpha_c, \beta_c)}{p(s)}, \quad (12)$$

the fitted distribution characterized by a larger parameter λ_c can be considered as representing the true negative distribution. This is particularly relevant in multi-class datasets, where true negatives are more prevalent than false negatives for each anchor class. Conversely, the fitted distribution with a smaller λ_c can be interpreted as the false negative distribution, given that both anchors and false negatives belong to the same class and typically exhibit closer similarity. The contribution of these negative samples to the loss function can then be computed using the posterior probabilities:

$$w(i, k) = \frac{p(c_t | s_{ik}) s_{ik}}{\frac{1}{B-1} \sum_{j \neq i} [p(c_t | s_{ij}) s_{ij}]}, \quad (13)$$

where s_{ik} is the similarity between anchor sample u_i and its negative sample v_k in the corresponding view, and $p(c_t | s_{ik})$ represents the probability that the negative sample v_k is a true negative relative to the anchor sample u_i . This probability can

be used for weighting both inter-view(u_i, v_k) and intra-view u_i, u_k samples:

$$l_w(u_i, v_i) = -\log \left(\frac{\phi(u_i, v_i)}{\sum w(i, k) \phi(u_i, v_k) + \sum_{k \neq i} w(i, k) \phi(u_i, u_k)} \right),$$

$$\phi(u, v) = \exp \left(\frac{\text{sim}(u, v)}{\tau} \right),$$
(14)

the overall contrastive loss after weighting is:

$$\mathcal{L}_W = \frac{1}{2N} \sum_{i=1}^N (l_w(u_i, v_i) + l_w(v_i, u_i))$$
(15)

Algorithm 1 CADGCL Algorithm for Unsupervised CAD Model Retrieval

Input: B-rep attribute graph $G^{(k)} = (V, E)$, initialized GNN encoder, epoch for BMM M

```

1: for epoch = 0, 1, 2, ... do
2:   Generate augmented view  $G'^{(k)} = (V, E')$  using Eq.(1) and Eq.(2)
3:   Encode the original graph  $G^{(k)}$  and augmented graph  $G'^{(k)}$  using Eq.(3)
4:   Project graph embeddings into a distinct vector space using Eq.(4)
5:   if epoch  $\geq M$  then
6:     Fit BMM parameters using EM algorithm
7:     Compute posterior probabilities using Eq. (12)
8:     Compute sample weights based on posterior probabilities using Eq.(13)
9:     Compute weighted contrastive loss  $L_W$  using Eq. (15)
10:    Minimize  $L_W$  to update model parameters
11:   else
12:     Compute  $L_C$  using Eq. (6)
13:     Minimize  $L_C$  to update model parameters
14:   end if
15: end for

```

Output: Optimized GNN encoder

4 Experiments

This section provides a detailed analysis of the application of the proposed CADGCL method in classifying and retrieving 3D CAD models. The discussion begins with an overview of the experimental setup, including descriptions of the dataset, evaluation metric, and training strategy. Following this, a systematic comparison is conducted between CADGCL and state-of-the-art models. The results of the retrieval tasks for the proposed method are then presented. Additionally, a comprehensive ablation study is performed to assess the significance of each module within the framework. Finally,



Fig. 3 Some model examples of the BRepCAD30 dataset

the impact of hyperparameters on model performance is analyzed.

4.1 Experiment preparation

Dataset: Among current CAD datasets, most publicly available datasets use point clouds or meshes as their primary representation formats. However, datasets based on B-rep remain relatively scarce. In contrast, B-rep can accurately describe CAD models' geometric shapes and topological structures and is widely used in industrial design and engineering manufacturing. To address this gap, a B-rep-based industrial parts dataset named BRepCAD30 is developed. The BRepCAD30 dataset is collaboratively developed with local enterprises and industrial partners and contains 10,365 parts, encompassing 30 types of CAD models with complex geometric shapes and diverse topological structures. These models reflect the diversity and complexity of part designs in real-world industrial scenarios. Examples of models in the dataset are shown in Fig. 3. In addition, to validate the effectiveness of the CADGCL method, experiments are conducted on the publicly available Fabwave[1] dataset. Fabwave is a publicly available B-rep-based industrial parts dataset widely used in the CAD field and considered a benchmark for evaluating CAD-related tasks. Due to its extensive adoption and authoritative status, Fabwave is a reliable reference for assessing the performance of CAD model retrieval and classification methods. This dataset comprises 4,475 CAD models across 45 categories, covering various industrial parts, including gears, flanges, and bolts.

Evaluation metric: Mean Average Precision (mAP) and F1 score. mAP is a commonly used metric for evaluating the performance of retrieval algorithms. It measures the accuracy of relevant results returned by the model during the retrieval process. Calculating mAP involves the following steps: First, for each query sample q , the Average Precision (AP) of its retrieval results is computed. Then, the map is obtained by taking the mean of the AP values across all queries.

F1 score is a commonly used metric for evaluating the performance of classification models, particularly in scenarios

involving imbalanced data. It provides a balanced assessment of model performance by combining Precision and Recall.

Training strategy: The dataset is partitioned into training and test subsets, adhering to an 80:20 split ratio. Specifically, 80% of the data is used for unsupervised training, where a GNN encoder is employed to learn CAD model embeddings under label-free conditions. The remaining 20% of the data is used for classification and retrieval tasks during the testing phase. A linear classifier is additionally trained for the classification task, using the embeddings generated from the test data as input. The classification performance is evaluated by calculating the F1-score. For the retrieval task, the embeddings from the test data are directly utilized to compute similarities and complete the retrieval process, with mAP as the evaluation metric to measure retrieval performance. The experimental settings are as follows: the GNN encoder's feature dimension is 256, with a feature masking rate of 0.3 and an edge perturbation rate of 0.1. The number of network layers is configured to 2; the model is trained using the Adam optimizer, with a learning rate of 0.01, a batch size of 128, and a total of 20 training epochs. All experiments used the PyTorch framework on an NVidia GeForce 4090 GPU.

4.2 Comparison to state-of-the-art approaches

The proposed CADGCL method is compared with current state-of-the-art methods, including two autoencoder-based point cloud methods (Point-MAE[17] and Point-FEMAE[30]) and seven graph contrastive learning methods (GraphCL[29], JOAO[28], SimGRACE[24], AutoGCL[27], GCLMI[26], CIGCL[21], and DRGCL[10]). As shown in Table 1, CADGCL achieved the best performance in both classification and retrieval tasks on the BRepCAD30 dataset. In the retrieval task, CADGCL achieved mAP scores of 84.51% for topk50 and 94.67% for topk10, significantly surpassing the performance of other methods. In the classification task, CADGCL achieved an F1 score of 94.67%, outperforming all comparison methods.

Compared to autoencoder-based point cloud methods, CADGCL significantly reduces computational resource requirements and delivers superior retrieval and classification task performance. Specifically, Point-MAE achieved an mAP (topk50) of 83.65%, and Point-FEMAE achieved 75.32%, while CADGCL outperformed them by 0.86 percentage points and 9.19 percentage points, respectively. These results demonstrate that the graph structure based on B-rep representation more effectively captures the geometric and topological information of CAD models, showcasing significant advantages in both performance and efficiency.

In comparison with graph contrastive learning methods, CADGCL also demonstrated significant advantages. For instance, the mAP (topk50) metric outperformed GraphCL and GCLMI by nearly three percentage points and four per-

centage points, respectively, while surpassing all comparison methods in the F1 score. The performance improvements of CADGCL can be attributed to two core innovations: the edge perturbation strategy based on EBC, which enhances data quality and semantic consistency by selectively perturbing semantically critical edges in the graph structure; the negative sampling strategy based on BMM, which optimizes the selection process of negative samples by modeling the similarity distribution of anchor samples to distinguish between false negatives and true negatives. These innovations enable CADGCL to achieve notable performance improvements over existing methods, demonstrating strong robustness and effectiveness.

To validate the generalization capability of the proposed CADGCL method, we evaluated its performance on the publicly available FabWave dataset. The results, shown in Table 2, demonstrate significant performance improvements in both retrieval and classification tasks. On the FabWave dataset, CADGCL achieved mAP (topk50) and mAP (topk10) scores of 89.35% and 98.58%, respectively, while the F1 score reached an impressive 98.84%, outperforming all comparison methods. Compared to existing methods, CADGCL's superior performance further highlights its robustness and generalization capability in complex CAD model retrieval and classification tasks. Compared with graph contrastive learning methods or autoencoder-based point cloud methods, CADGCL exhibited exceptional generalization on publicly available datasets.

To further illustrate the performance of CADGCL in CAD model retrieval tasks, a visualization analysis of its retrieval results is performed using the BRepCAD30 dataset. After completion of model training, embeddings are generated from the best-saved model. For the retrieval task, 10 CAD models are randomly selected from the test set as query samples. The top 10 entity models with the highest cosine similarity are retrieved from the embedding space. Figure 4 presents the detailed retrieval results: the first column of each row represents the query sample. In contrast, the remaining columns display the results retrieved by the CADGCL method, ranked from left to right based on their distance in the latent space. The green-boxed results highlight retrieval diversity, showcasing models maintaining key geometric and topological similarities to the query while exhibiting distinct structural variations. Specifically, for the Gear model in the first row, the second green-boxed retrieval result introduces a central boss with an axial step transition, accompanied by a material reduction on the opposite side. Similarly, for the Nut model in the second row, the second green-boxed retrieval result exhibits a decreased axial extrusion, a thin-walled profile transformation, and an omission of internal threading. These retrieved variants are not merely geometric modifications but also reflect potential differences in design intent—such as adaptation to varying load condi-

Table 1 Retrieval and classification results on the BRepCAD30 dataset. The best performance is highlighted in **bold**, while the second-best performance is indicated with underlining

Approach	Retrieval mAP(top50) (%)	mAP(top10) (%)	Classification F1 (%)
GraphCL [29]	81.25	91.89	93.30
JOAO [28]	82.18	92.35	93.32
SimGRACE [24]	81.31	<u>93.18</u>	91.39
AutoGCL [27]	79.09	90.64	92.74
GCLMI [26]	79.88	91.10	<u>93.53</u>
CIGCL [21]	73.83	88.99	88.49
DRGCL [10]	78.09	90.08	92.89
Point-MAE [17]	<u>83.65</u>	92.52	93.27
Point-FEMAE [30]	75.32	87.22	92.89
CADGCL (ours)	84.51	94.67	94.67

Table 2 Retrieval and classification results on the Fabwave dataset. The best performance is highlighted in **bold**, while the second-best performance is indicated with underlining

Approach	Retrieval mAP(tok50) (%)	mAP(topk10) (%)	Classification F1 (%)
GraphCL [29]	72.72	92.92	94.76
JOAO [28]	84.64	97.01	97.32
SimGRACE [24]	82.78	93.22	94.54
AutoGCL [27]	85.82	96.72	94.68
GCLMI [26]	83.34	96.98	96.03
CIGCL [21]	85.63	96.96	95.87
DRGCL [10]	<u>86.18</u>	<u>97.13</u>	<u>96.71</u>
Point-MAE [17]	84.44	94.30	95.24
Point-FEMAE [30]	83.82	94.18	94.86
CADGCL (ours)	89.35	98.58	98.84

tions, assembly requirements, or manufacturing constraints. In practical industrial workflows, such retrieval capabilities enable engineers to identify existing components that fulfill similar functional roles quickly while exhibiting structural variations. This facilitates efficient design reuse and eliminates redundant modeling efforts. By enabling the retrieval of structurally diverse yet functionally similar components without supervision, CADGCL serves as a practical tool for streamlining design processes and supporting informed engineering decisions in CAD-intensive environments.

4.3 Ablation experiment

To comprehensively evaluate the impact of each module in CADGCL, an ablation study is conducted, and the results are presented in the Table 3. The ablation study analyzed the impact of the two proposed modules on model performance:

- *w/o EBC & BMM*: Both the EBC-based edge perturbation strategy and the BMM-based negative sampling strategy are removed.
- *w/o EBC*: Only the EBC-based edge perturbation strategy is removed.

- *w/o BMM*: Only the BMM-based negative sampling strategy is removed.
- *CADGCL*: The EBC-based edge perturbation strategy and the BMM-based negative sampling strategy are used, constituting our model's final architecture.

The results show that CADGCL performs best when all modules are retained. Compared to the ablation experiments where key modules are removed, CADGCL demonstrated significant performance improvements across all metrics, confirming each module's critical role in enhancing the model's overall performance.

4.4 Hyperparameter analysis

To understand how key hyperparameters influence model behavior, we conduct an ablation study on the feature masking ratio and edge perturbation ratio, with the results shown in Fig. 5. The model maintained high performance at low masking ratios and initially benefited from increased masking, reaching peak performance at a ratio of 0.3. This moderate masking encourages the model to focus on essential geometric and topological features by suppressing redundant

Fig. 4 Retrieval results on BRepCAD30 dataset. The first column is the query, other columns are retrieved results sorted left to right by distance in latent space

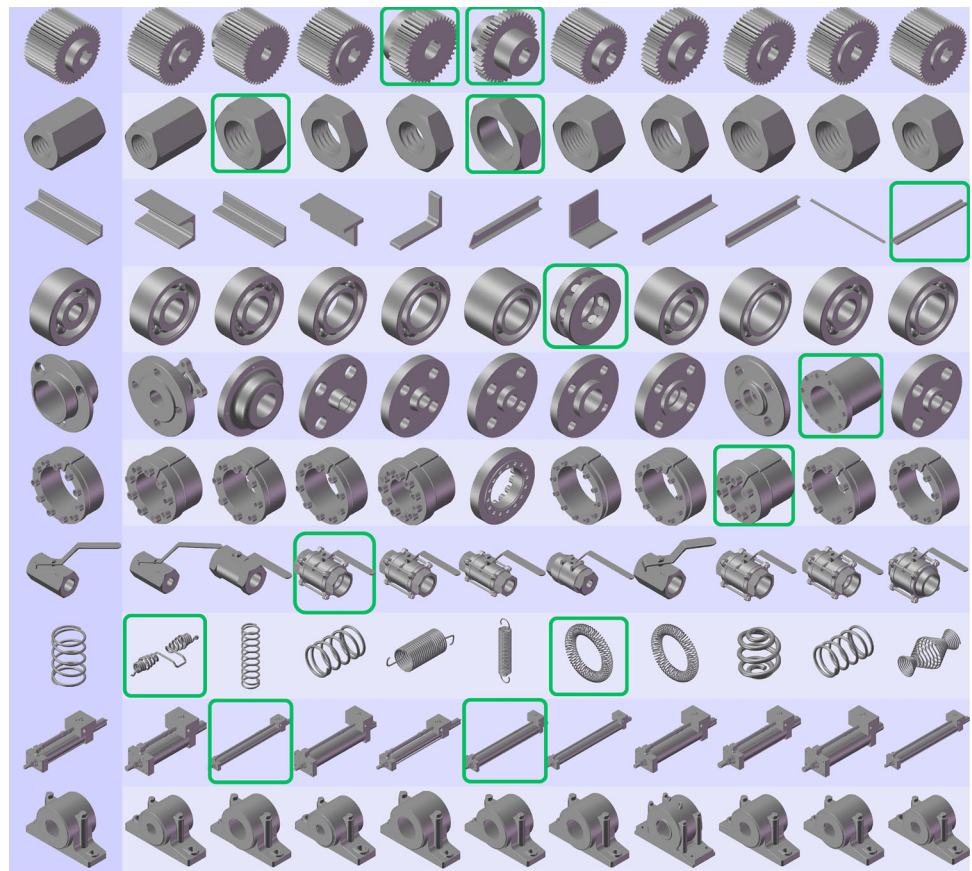


Table 3 Effectiveness of different components of CADGCL

Ablation study	mAP (topk50) (%)	mAP (topk10) (%)	F1 (%)
<i>w/o EBC & BMM</i>	78.32	90.51	91.84
<i>w/o EBC</i>	80.18	91.28	92.11
<i>w/o BMM</i>	81.26	92.27	92.97
CADGCL	84.51	94.67	94.67

Bold font is used to highlight the configuration that achieves the best performance among all ablation settings

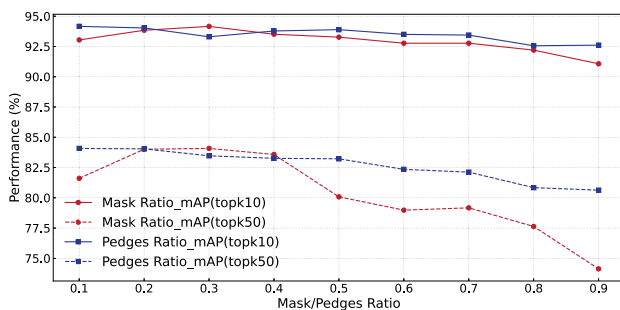


Fig. 5 Ablation study on the effects of mask ratio and pledges ratio on model performance

information, thereby enhancing representation robustness. However, as the masking ratio increased to 0.9, performance dropped significantly. This degradation is likely due to

excessive masking disrupting the integrity of input features, leading to insufficient information for consistent representation learning. Similarly, model performance peaked at a perturbation ratio of 0.1 but declined as the ratio increased, with a sharp drop observed beyond 0.5. This suggests that moderate edge perturbation enhances model robustness, whereas excessive perturbation disrupts the graph structure, impairing the capture of geometric and topological details. These results further validate the importance of proper hyper-parameter selection in enhancing model performance and robustness.

4.5 Visualization analysis

To illustrate how different methods perform in the latent feature space of 3D CAD models, the t-SNE visualizations are conducted on the test set. The t-SNE algorithm

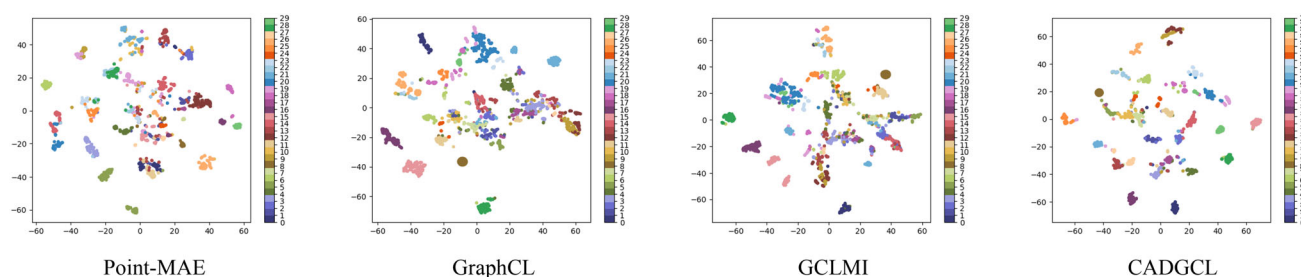


Fig. 6 T-SNE visualization of the latent feature space

projects high-dimensional features into a two-dimensional space, allowing us to observe the clustering behavior of different categories. As shown in Fig. 6, the visualization results reveal significant differences in feature distributions across various methods. CADGCL effectively maps 3D CAD models into the latent feature space, forming compact category clusters where intra-class samples are closely grouped. At the same time, inter-class boundaries remain well-defined, demonstrating strong discriminative capability. This indicates that CADGCL learns more distinctive feature representations, enhancing performance in 3D CAD model retrieval and classification tasks. Overall, CADGCL exhibits the best performance in the feature space, with the most compact intra-class clustering, clear category boundaries, and minimal inter-class overlap, further validating its effectiveness in 3D CAD model retrieval.

5 Conclusion

This study introduces CADGCL, an unsupervised method for retrieving CAD models, aimed at overcoming the limitations of conventional retrieval approaches that heavily depend on extensive labeled datasets. The method converts CAD models into B-rep attributed graphs and successfully learns graph representations of geometric and topological structures using a graph contrastive learning framework, enabling efficient retrieval and classification. To resolve semantic information loss and sampling bias issues in traditional graph contrastive learning, we introduce the edge perturbation strategy based on EBC and the negative sample selection strategy based on BMM. The EBC strategy improves the quality and consistency of data augmentation by preserving the semantic information of critical edges. In contrast, the BMM strategy significantly reduces the interference of false negative samples by modeling the similarity distribution of negative samples. Experimental results demonstrate that CADGCL achieves significant performance improvements in retrieval and classification tasks on the BRepCAD30 and FabWave datasets, fully validating its effectiveness and generalization capability in unsupervised CAD model retrieval tasks.

Future research will further explore the potential extensions of CADGCL. On the one hand, combining graph contrastive learning with graph autoencoders (e.g., Graph MAE) could enhance the model's feature representation capability by introducing reconstruction tasks. On the other hand, integrating multimodal data (e.g., point clouds and B-rep representations) could improve the model's understanding and adaptability to complex geometric and topological features of CAD models. These directions offer new perspectives for optimizing CADGCL's performance and expanding its applications in various scenarios.

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Author Contributions F.Q. proposed the research idea, supervised the project, and provided overall guidance and manuscript direction. L.Z. designed and implemented the proposed method, carried out the experiments, and wrote the manuscript. Z.X. assisted in diagram creation, conducted comparative experiments, and participated in manuscript editing. M.F. and P.L. provided writing guidance and contributed to manuscript revisions. All authors reviewed and approved the manuscript.

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Data Availability Statement No datasets were generated or analysed during the current study

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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References

- Angrish, A., Craver, B., Starly, B.: "FabSearch": A 3D CAD model-based search engine for sourcing manufacturing services. *J. Comput. Inf. Sci. Eng.* **19**(4), 041,006 (2019)
- Dai, J., Fan, R., Song, Y., Guo, Q., He, F.: MEAN: an attention-based approach for 3D mesh shape classification. *Vis. Comput.* **40**, 2987–3000 (2024)
- Dempster, A.P., Laird, N.M., Rubin, D.B.: Maximum likelihood from incomplete data via the EM algorithm. *J. R. Stat. Soc.: Series B (Methodological)* **39**(1), 1–22 (2018)
- DP10303, S.: Standard for the exchange of product model data. Tech. rep., ISO TC184/SC4/WG1 (1988)
- Girvan, M., Newman, M.E.: Community structure in social and biological networks. *Proceed. Natl. Acad. Sci.* **99**(12), 7821–7826 (2002)
- Gupta, A.K., Nadarajah, S.: *Handbook of Beta Distribution and Its Applications*. CRC Press, Boca Raton (2004)
- Hanocka, R., Hertz, A., Fish, N., Giryas, R., Fleishman, S., Cohen-Or, D.: MeshCNN: a network with an edge. *ACM Trans. Graphics (ToG)* **38**(4), 1–12 (2019)
- Hou, J., Luo, C., Qin, F., Shao, Y., Chen, X.: FuS-GCN: efficient B-rep based graph convolutional networks for 3D-CAD model classification and retrieval. *Adv. Eng. Inf.* **56**, 102,008 (2023)
- Jayaraman, P.K., Sanghi, A., Lambourne, J.G., Willis, K.D.D., Davies, T., Shayani, H., Morris, N.: UV-Net: learning from Boundary Representations. In: 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 11,698–11,707 (2021)
- Ji, Q., Li, J., Hu, J., Wang, R., Zheng, C., Xu, F.: Rethinking dimensional rationale in graph contrastive learning from causal perspective. In: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 12,810–12,820 (2024)
- Ji, Y., Wu, C., Liu, P., Wang, J., Coombes, K.R.: Applications of beta-mixture models in bioinformatics. *Bioinformatics* **21**(9), 2118–2122 (2005)
- Kim, H., Yeo, C., Cha, M., Mun, D.: A method of generating depth images for view-based shape retrieval of 3D CAD models from partial point clouds. *Multimed. Tools Appl.* **80**, 10859–10880 (2021)
- Lee, J., Yeo, C., Cheon, S.U., Park, J.H., Mun, D.: BRepGAT: graph neural network to segment machining feature faces in a B-rep model. *J. Comput. Design Eng.* **10**(6), 2384–2400 (2023)
- Liang, Y., He, F., Fan, B., Tang, W.: MeshCL: towards robust 3D mesh analysis via contrastive learning. *Adv. Eng. Inf.* **60**, 102,439 (2024)
- Maturana, D., Scherer, S.: VoxNet: a 3D convolutional neural network for real-time object recognition. In: 2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 922–928. IEEE (2015)
- Oord, A.v.d., Li, Y., Vinyals, O.: Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748* (2018)
- Pang, Y., Wang, W., Tay, F.E.H., Liu, W., Tian, Y., Yuan, L.: Masked autoencoders for point cloud self-supervised learning. In: European Conference on Computer Vision, pp. 604–621. Springer (2022)
- Qi, C.R., Su, H., Mo, K., Guibas, L.J.: PointNet: Deep learning on point sets for 3D classification and segmentation. In: IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 77–85 (2017)
- Qi, C.R., Yi, L., Su, H., Guibas, L.J.: PointNet++: Deep hierarchical feature learning on point sets in a metric space. In: Advances in Neural Information Processing Systems (2017)
- Qin, F., Zhan, G., Fang, M., Chen, C.P., Li, P.: VGNet: Multimodal feature extraction and fusion network for 3D CAD model retrieval. *IEEE Transactions on Multimedia* pp. 1–16 (2024)
- Tan, S., Li, D., Jiang, R., Zhang, Y., Okumura, M.: Community-invariant graph contrastive learning. *arXiv preprint arXiv:2405.01350* (2024)
- Wang, Y., Sun, Y., Liu, Z., Sarma, S.E., Bronstein, M.M., Solomon, J.M.: Dynamic graph CNN for learning on point clouds. *ACM Trans. Graphics (tog)* **38**(5), 1–12 (2019)
- Wu, Z., Song, S., Khosla, A., Yu, F., Zhang, L., Tang, X., Xiao, J.: 3D ShapeNets: a deep representation for volumetric shapes. In: Proceeding of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 1912–1920 (2015)
- Xia, J., Wu, L., Chen, J., Hu, B., Li, S.Z.: SimGRACE: a simple framework for graph contrastive learning without data augmentation. In: Proceedings of the ACM Web Conference 2022, pp. 1070–1079 (2022)
- Xia, J., Wu, L., Wang, G., Chen, J., Li, S.Z.: ProGCL: rethinking hard negative mining in graph contrastive learning. In: Proceedings of the 39th International Conference on Machine Learning, pp. 24,332–24,346. PMLR (2022)
- Xu, Y., Wang, J., Guang, M., Yan, C., Jiang, C.: Graph contrastive learning with min-max mutual information. *Inf. Sci.* **665**, 120,378 (2024)
- Yin, Y., Wang, Q., Huang, S., Xiong, H., Zhang, X.: AutoGCL: automated graph contrastive learning via learnable view generators. In: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 8892–8900 (2022)
- You, Y., Chen, T., Shen, Y., Wang, Z.: Graph contrastive learning automated. In: Proceedings of the 38th International Conference on Machine Learning, pp. 12,121–12,132. PMLR (2021)
- You, Y., Chen, T., Sui, Y., Chen, T., Wang, Z., Shen, Y.: Graph contrastive learning with augmentations. In: Proceedings of the 34th International Conference on Neural Information Processing Systems, pp. 5812–5823 (2020)
- Zha, Y., Ji, H., Li, J., Li, R., Dai, T., Chen, B., Wang, Z., Xia, S.T.: Towards compact 3D representations via point feature enhancement masked autoencoders. In: Proceedings of the AAAI Conference on Artificial Intelligence, pp. 6962–6970 (2024)
- Zhang, D., He, F., Tu, Z., Zou, L., Chen, Y.: Pointwise geometric and semantic learning network on 3D point clouds. *Integrated Comput.-Aided Eng.* **27**(1), 57–75 (2020)
- Zhou, H., Wang, F., Liu, Q., Li, J., Liu, W., Li, X., Liu, A.A.: Universal unsupervised cross-domain 3D shape retrieval. *Multimed. Syst.* **30**, 17 (2024)
- Zhu, Y., Xu, Y., Liu, Q., Wu, S.: An empirical study of graph contrastive learning. *arXiv preprint arXiv:2109.01116* (2021)

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